

DATA ANALYST INTERNSHIP

Task 2: Data Visualization and Storytelling

Clear Task Solution for GitHub Submission

1. Objective

Create visualizations that communicate a clear business story from sales data. The analysis below is structured so it can be implemented in Tableau or Power BI using the Superstore dataset.

2. Dataset

Item	Details
Dataset	Superstore.csv / sales dataset
Tools	Tableau or Power BI
Main analysis	Sales, profit, category, region, time and customer segments
Output	Visual report / dashboard screenshots in PDF

3. Dashboard / Visual Report Design

Visual	Recommended chart	Business purpose
KPI Summary	Cards	Show total sales, total profit, orders and profit margin
Sales Trend	Line chart	Identify monthly/yearly sales movement
Sales by Category	Bar chart	Compare category performance
Profit by Category	Bar chart	Identify profitable and weak categories
Regional Performance	Column/bar chart or map	Compare regions geographically
Segment Analysis	Bar chart	Compare customer segment contribution
Sales vs Profit	Scatter plot	Identify high-sales/low-profit areas

4. Required Visualizations

1. KPI Cards: Display Total Sales, Total Profit, Number of Orders and Profit Margin.

2. Sales Trend: Use a line chart with Date on the X-axis and Sales on the Y-axis. Add a time filter if available.

3. Category Comparison: Use a horizontal bar chart with Category on the axis and Sales as the value. Sort from highest to lowest.

4. Profit Analysis: Use a bar chart for Profit by Category or Sub-Category. This helps identify products contributing to or reducing profitability.

5. Regional Analysis: Compare Sales and Profit across regions. A map can be used if geographic fields are available.

6. Customer Segment: Compare Sales and Profit for each customer segment.

7. Sales vs Profit: Use a scatter plot to identify combinations of high sales and high/low profit.

5. Dashboard Story

Start with the KPI cards to establish the overall business position. Next, use the sales trend to show how performance changes over time. Category and sub-category charts explain where sales and profit are being generated. Regional and segment analysis then identify the strongest markets and customer groups. Finally, the sales-versus-profit view helps management identify areas where high sales may not translate into strong profitability.

6. Business Insights to Report

- Identify the category with the highest sales.
- Identify the category or sub-category with the highest profit.
- Identify categories or sub-categories with comparatively weak or negative profit.
- Identify the region contributing the most sales and profit.
- Identify the customer segment contributing the highest sales.
- Identify months or periods with unusually high or low sales.
- Look for high-sales areas with low profit, which may indicate discounting, high costs or an unfavorable product mix.

7. Visualization Best Practices Applied

Practice	Implementation
Choose the right chart	Use line charts for trends, bars for comparisons, cards for KPIs and scatter plots for relationships.
Reduce clutter	Remove unnecessary gridlines, labels and decorative elements.
Use limited colors	Use a consistent visual theme and highlight only important values.
Add context	Give every chart a meaningful title and, where appropriate, data labels or annotations.
Focus on insights	Explain what the chart means for the business rather than presenting visuals alone.
Create a summary	End the dashboard/report with key findings and recommended areas for attention.

8. Final Summary

The completed visualization report should provide a single, easy-to-understand view of sales performance. The dashboard combines KPIs, trends, category performance, profitability, regional performance and customer segments to turn raw sales data into actionable business insights. The final output can be exported as a PDF or submitted using dashboard screenshots.

9. GitHub Submission Structure

Task-2-Data-Visualization/

■■■ README.md

■■■ dataset/

■ ■■■ Superstore.csv

■■■ dashboard/

- ■■■ dashboard_screenshot_1.png
- ■■■ dashboard_screenshot_2.png
- report/
- Task_2_Data_Visualization_Solution.pdf

Note: The interview questions and the task's submission guidelines are intentionally excluded, as requested. This document contains only the task solution, visualization plan, business-story structure and GitHub-ready organization.